FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

1

RANDOM WALK SIMULATION USING AGENTPY

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I. BACKGROUND INFORMATION

A **random walk** is a mathematical process that describes a path consisting of a sequence of **random steps**. Random walks are used in many fields, such as physics, biology, economics, and computer science, to model systems that evolve randomly over time.

In this laboratory, we simulate multiple agents moving randomly on a grid using **AgentPy**, a Python library for **agent-based modeling**. Each agent represents an entity that can move one step at a time in a random direction. This simulation helps students understand:

- How **agents move and interact** in a grid-based environment.
- The concept of **bounded movement**, ensuring agents stay within the grid limits.
- What changing **parameters** like the number of agents, grid size, or number of steps affects the system.
- How **agent-based models** can be used to simulate real-world phenomena, such as **diffusion, crowd dynamics, or swarm behavior**.

II. TECHNICAL ASSESSMENT

RANDOM WALK SIMULATION USING AGENTPY

Objective

At the end of this laboratory activity, students are expected to:

1. To understand the concept of **random walks** in a grid environment.
2. To implement a **multi-agent simulation** using Python and AgentPy.
3. To visualize the **movement of agents** on a grid over time.

Materials Needed

- Computer with Python 3.x installed
- Libraries: agentpy, matplotlib
- Code editor (e.g., VSCode, PyCharm, or Jupyter Notebook)

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs in the Command Window.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA1_lastname.pdf**

(Example: TA1_Patilano.pdf)

8. Submit the **python file** and **PDF** file.

Activity Instructions

Step 1: Run the Code

1. Open the provided Python code.

Random Walk Simulation (Console-Based)

```
import agentpy as ap
import random
import matplotlib.pyplot as plt
import matplotlib.animation as animation

# -------- Get user input from console --------
num_agents = int(input("Enter number of agents: "))
grid_size = int(input("Enter grid size (e.g., 10 for 10x10): "))
num_steps = int(input("Enter number of steps: "))

# -------- Agent Definition -----------
class RandomWalker(ap.Agent):
    def step(self):
        direction = random.choice([(1,0), (-1,0), (0,1), (0,-1)])
        x, y = self.position
        # Keep agents inside grid
        x = max(0, min(self.model.p.grid_size[0]-1, x + direction[0]))
        y = max(0, min(self.model.p.grid_size[1]-1, y + direction[1]))
        self.position = (x, y)

# -------- Model Definition -----------
class RandomWalkModel(ap.Model):
    def setup(self):
        # Create agents
        self.agents = ap.AgentList(self, self.p.agents, RandomWalker)

        # Initialize random positions
        for agent in self.agents:
            agent.position = (random.randint(0, self.p.grid_size[0]-1),
                              random.randint(0, self.p.grid_size[1]-1))

        # Create grid (optional)
        self.grid = ap.Grid(self, self.p.grid_size, torus=False)
        self.grid.add_agents(self.agents)

    def step(self):
        for agent in self.agents:
            agent.step()

# -------- Parameters from user input -----------
parameters = {
    'agents': num_agents,
    'grid_size': (grid_size, grid_size),
    'steps': num_steps
```

```
}

# -------- Run Model ----------
model = RandomWalkModel(parameters)
model.setup()

# -------- Interactive Animation ----------
fig, ax = plt.subplots()
ax.set_xlim(0, model.p.grid_size[0])
ax.set_ylim(0, model.p.grid_size[1])
ax.set_xticks(range(model.p.grid_size[0]+1))
ax.set_yticks(range(model.p.grid_size[1]+1))
ax.grid(True)

scat = ax.scatter([], [], s=200)  # s = size of agents

def update(frame):
    model.step()
    x = [agent.position[0] for agent in model.agents]
    y = [agent.position[1] for agent in model.agents]
    scat.set_offsets(list(zip(x, y)))
    return scat,

ani = animation.FuncAnimation(fig, update, frames=model.p.steps, blit=True, repeat=False)
plt.show()
```

2. Run the program in your terminal.
3. Enter the following values when prompted:
    - **Number of agents** (e.g., 5)
    - **Grid size** (e.g., 10 for a 10x10 grid)
    - **Number of steps** (e.g., 20)

These values control the number of agents, the size of the environment, and the simulation duration.

Step 2: Understand the Agent Behavior

1. Each agent is represented by a dot in the grid.
2. Agents move **randomly** in one of four directions: up, down, left, or right.
3. Agents **cannot leave the grid**; movement is restricted within the grid boundaries.

Step 3: Observe the Simulation

1. After running, a **grid with moving agents** is displayed.
2. Each time step updates the position of all agents.
3. Observe:
    - How agents spread across the grid over time
    - Whether agents cluster in certain areas or move freely

Step 4: Experiment with Parameters

- Repeat the simulation with **different values**:

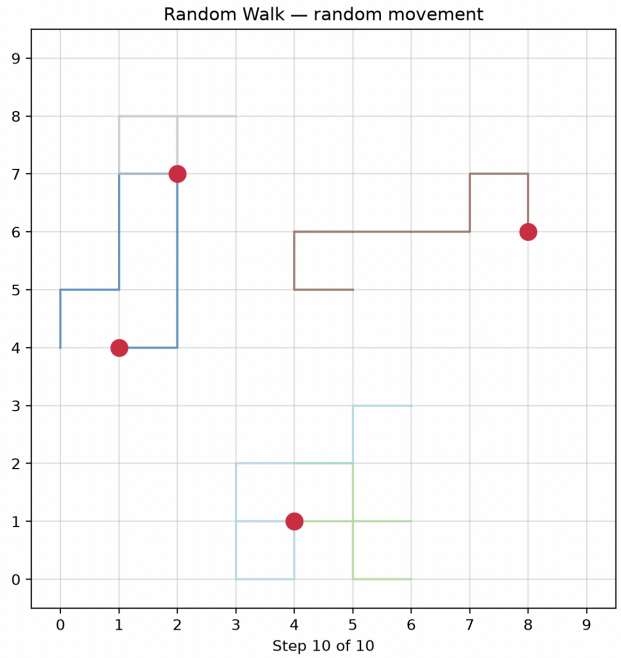
- **More agents** (e.g., 10 or 20)
- **Smaller/larger grid sizes**
- **Longer simulations** (e.g., 50 steps)

- Observe and record how the **agent movement patterns change**.

Laboratory Tasks

1. **Track agent paths**: Modify the code to **record and display the path of each agent** during the simulation.
2. **Change movement rules**: Make agents **avoid each other** or move in a **preferred direction** and observe the resulting patterns.
3. **Analyze final positions**: Record the **final positions** of all agents and analyze their **distribution**.
4. **Compare scenarios**: Run **two simulations with different parameters** and compare the movement patterns.

Modified (Updated Code - Laboratory Tasks) ⇩



```
import random
import agentpy as ap
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation
```

```
class RandomWalker(ap.Agent):
    def step(self):
        x, y = self.position
        moves = [(1, 0), (-1, 0), (0, 1), (0, -1)]

        # Make rightward movement more likely when selected.
        if self.model.p.movement_rule == "preferred":
            moves += [(1, 0), (1, 0)]

        # Avoid cells occupied by other agents when possible.
        if self.model.p.movement_rule == "avoid":
            occupied = {
                other.position
                for other in self.model.agents
                if other is not self
            }
            safe_moves = [
                move for move in moves
                if self.model.bounded_position(
                    x + move[0], y + move[1]
                ) not in occupied
            ]
            if safe_moves:
                moves = safe_moves

        dx, dy = random.choice(moves)
        self.position = self.model.bounded_position(x + dx, y + dy)
        self.path.append(self.position)


class RandomWalkModel(ap.Model):
    def bounded_position(self, x, y):
        """Keep coordinates inside the grid."""
        maximum = self.p.grid_size - 1
        return (
            max(0, min(maximum, x)),
            max(0, min(maximum, y)),
        )
```

```
    def setup(self):
        self.agents = ap.AgentList(self, self.p.agents, RandomWalker)

        for agent in self.agents:
            agent.position = (
                random.randint(0, self.p.grid_size - 1),
                random.randint(0, self.p.grid_size - 1),
            )
            agent.path = [agent.position]

    def step(self):
        for agent in self.agents:
            agent.step()


def get_positive_int(prompt):
    while True:
        try:
            value = int(input(prompt))
            if value > 0:
                return value
        except ValueError:
            pass
        print("Please enter a whole number greater than zero.")


def main():
    print("\n--- Random Walk Simulation ---")
    num_agents = get_positive_int("Enter number of agents: ")
    grid_size = get_positive_int("Enter grid size: ")
    num_steps = get_positive_int("Enter number of steps: ")

    rule = input(
        "Movement rule (random / avoid / preferred) "
        "[default: random]: "
    ).strip().lower()

    if rule not in {"random", "avoid", "preferred"}:
```

```
    rule = "random"

parameters = {
    "agents": num_agents,
    "grid_size": grid_size,
    "steps": num_steps,
    "movement_rule": rule,
}

model = RandomWalkModel(parameters)
model.setup()

print("\nInitial positions:")
for number, agent in enumerate(model.agents, start=1):
    print(f"Agent {number}: {agent.position}")

fig, ax = plt.subplots(figsize=(7, 7))
ax.set_title(f"Random Walk — {rule} movement")
ax.set_xlim(-0.5, grid_size - 0.5)
ax.set_ylim(-0.5, grid_size - 0.5)
ax.set_xticks(range(grid_size))
ax.set_yticks(range(grid_size))
ax.grid(True, alpha=0.4)

positions = [agent.position for agent in model.agents]
dots = ax.scatter(
    [position[0] for position in positions],
    [position[1] for position in positions],
    s=120,
    color="crimson",
    zorder=3,
)

colors = plt.colormaps["tab20"].resampled(num_agents)
path_lines = [
    ax.plot([], [], color=colors(i), linewidth=1.5, alpha=0.7)[0]
    for i in range(num_agents)
]
```

```
last_frame = -1

def update(frame):
    nonlocal last_frame

    # Matplotlib may request the first frame more than once.
    if frame > last_frame:
        model.step()
        last_frame = frame

    positions = [agent.position for agent in model.agents]
    dots.set_offsets(positions)

    for agent, line in zip(model.agents, path_lines):
        x_values, y_values = zip(*agent.path)
        line.set_data(x_values, y_values)

    ax.set_xlabel(f"Step {last_frame + 1} of {num_steps}")
    return [dots, *path_lines]

# Keep this variable alive while the window is open.
animation = FuncAnimation(
    fig,
    update,
    frames=num_steps,
    interval=300,
    repeat=False,
    blit=False,
)

plt.show()

print("\nFinal positions and recorded paths:")
for number, agent in enumerate(model.agents, start=1):
    print(f"Agent {number}:")
    print(f"  Final position: {agent.position}")
    print(f"  Path: {agent.path}")
```

```
if __name__ == "__main__":
    main()
```

Step 5: Answer the Questions

1. What happens if the number of agents is **very high** compared to the grid size?
2. How does increasing the number of steps affect the **spread of agents**?
3. Do the agents ever move outside the grid? Why or why not?
4. How can this simulation be used to **model real-life phenomena** (e.g., diffusion, crowd movement)?
5.

Answer the Questions ⇩

1. The grid becomes crowded, so agents may cluster or share the same cell.
2. More steps let agents explore more of the grid, though they may revisit cells.
3. No. The code keeps every position within the grid boundaries.
4. It can show how particles spread (diffusion) or how people move through a crowded space.